TEST CODE **02138032**

FORM TP 2007-SPEC

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

ADVANCED PROFICIENCY EXAMINATION

PHYSICS - SPECIMEN PAPER

UNIT 1 - PAPER 032

ALTERNATIVE TO SBA

2 hours

Candidates are advised to use the first 15 minutes for reading through this paper carefully.

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of THREE questions. Attempt ALL questions.

2. The use of silent non-programmable calculators is allowed.

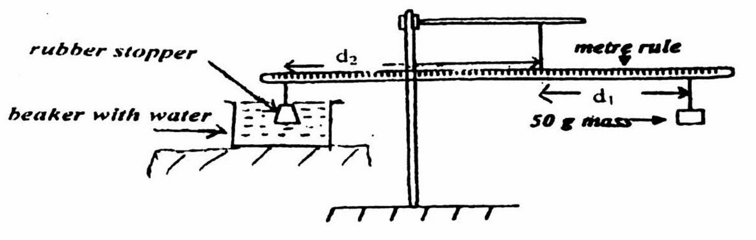
Question 1

The apparatus shown in Figure 1 is used to determine the volume and the mass of the rubber stopper using the upthrust.

Method:

(a) First adjust the position of the thread on the rule so that it balances horizontally on its own with no other masses suspended. Record the position of the thread.

Figure 1



Take the rubber stopper provided and suspend the thread close to one end of the metre rule. Now balance the rule by suspending a 100 g mass by a thread close to the other side of the rule. The rule should be horizontal when balanced. Record the distances d_1 and d_2 respectively and the density of the fluid in the beaker in Table 1 below.

Table 1

Fluid	Water	Paraffin Oil	Air	Methylated Spirit	Turpentine	Alcohol
Density (ρ) g/cm^3						
Distance (d_1) cm						

Repeat the procedure after replacing the fluid in the beaker with the following fluids to complete Table 1.

1. Paraffin oil
2. Air
3. Methylated spirit
4. Turpentine
5. Alcohol

6 marks

(b) When the rule is balanced the principal of moments states that the sum of the moments of the forces about the point of suspension in the clockwise direction is equal to the sum of the moments in the anticlockwise direction. The equation representing the principle of moments is as follows.

$$50 \times g \times d_1 = \text{Upthrust} \times d_2$$

$$50 \times g \times d_1 = (\rho V g \; m_s g) \times d_2$$

Rearranging

$$\rho = \frac{50\, d_1}{V\, d_2} - \frac{m_s}{V}$$

Plot a graph of density (ρ) versus distance (d_1) on the grid on page 3.

5 marks

(c) Determine the slope (S) of the graph you have drawn.

3 marks

(d) Use the value of the slope (S) to calculate the volume V of the stopper.

2 marks

(e) Use the graph to find the intercept (C) on the density axis.

1 mark

(f) Use the value of the intercept to find the mass of the stopper.

2 marks

Total 19 marks

Question 2

Given the following apparatus investigate the variation of light intensity with the thickness of glass. Each glass side is covered with automotive tint.

Apparatus: Glass slides 5 inches x 5 inches
Light intensity meter
Light source
Meter rule
Matt black card
Micrometer screw gauge.

Identification of variables:

____

____

3 marks

Procedure:

4 marks

Results:

3 marks

Conclusion:

2 marks

Total 12 marks

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Question 3

Table 2 represents the data obtained from a Young Modulus experiment.

Table 2.

Force (F) /N	1.9	6.08	9.12	9.88	11.59	14.06
Extension(e)/m	0.01	0.032	0.048	0.052	0.0.61	0.074

The original length of the wire used in the experiment was 1.85 m and the diameter of the wire was 2.5 mm.

The equation for then Young Modulus E of the wire is $E = \frac{F/A}{e/L}$

F is the force on the wire, A is the cross sectional area of the wire, e is the extension and L is the original length of the wire.

(a) On the graph page on page 8 plot a graph of force versus extension.

7 marks

(b) Use the graph to determine the young Modulus of the wire.

10 marks

Total 17 marks

END OF TEST